

Course: Domain Driven Design & Microservices for Architects

Section: Understanding the Domain

<http://acloudfan.com/>

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Discount Link to course:

<https://www.udemy.com/course/domain-driven-design-and-microservices/?referralCode=C5DCD3C4CC0F0298EC1A>

Understanding the Domain

Terminology

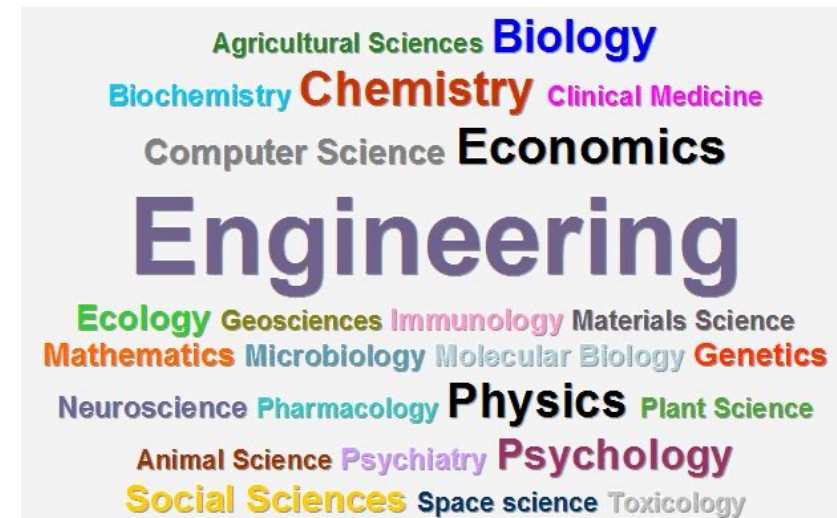


- 1 What is a domain?
- 2 Domain experts?
- 3 Intro to ACME Travels

English Language definition of *Domain*



A sphere of knowledge, influence, or activity



Business perspective of *Domain*



A field | industry in which the business operate

Banking



Bank of America

Oil & Gas



Retail

Walmart 



Flipkart

... ..

Technology perspective of *Domain*



Represents the problem space

eCommerce

eCommerce

User Experience | Front End

Business
Logic | Rules | Flow

Data
Layer

Gateway

Adapters

Interfaces

Software perspective of *Domain*



Represents the problem space

eCommerce

Social Media

Media
Streaming

Resource
Planning

www.AClou
amazon.com
m

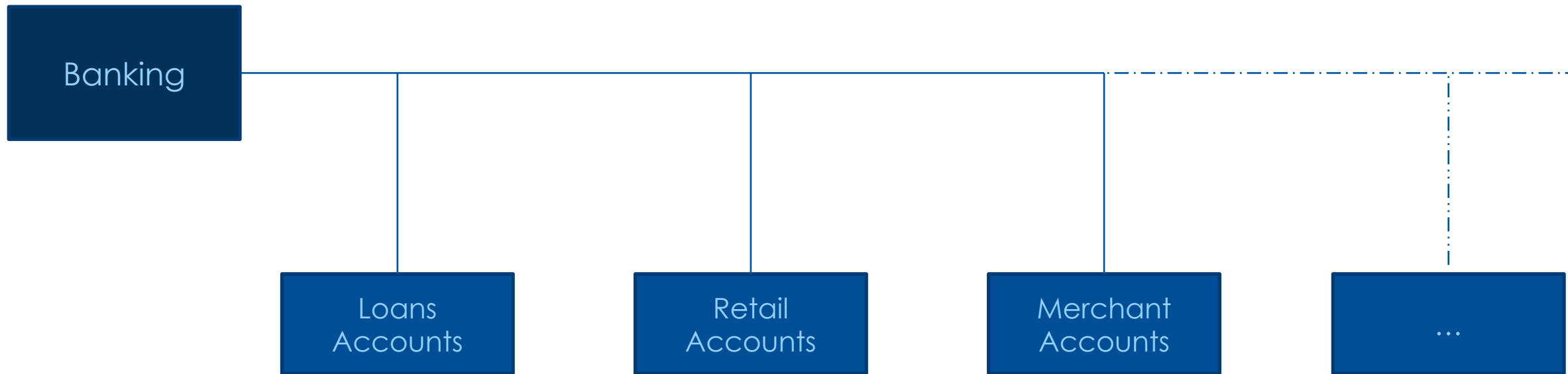


NETFLIX



Sub-Domains

Each domain consists of Sub-Domains



Domain Experts

a.k.a. Subject Matter Experts or SME

Thorough understanding of the domain

Banking



Accounts Expert

Domain Experts

a.k.a. Subject Matter Experts or SME

NO one expert knows everything about the domain !!!

Banking



Retail Accounts Expert



Merchant Accounts Expert



Loans Accounts Expert



Compliance & Regulatory

Travel & Leisure industry



Travel Advisor

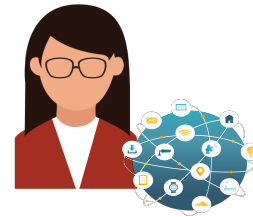




There are multiple experts within a domain !!!



Travel Advisor



Partner Contracts



Accounts



Customer Support

... ..

Quick Exercise



What domain are you in?



Subdomains within that domain?



List out the domain experts you work with?



Quick Review

Domain = A sphere of knowledge, influence, or activity

- Made up of MULTIPLE Sub-domains
- Multiple Domain Experts needed to support business functions

Architecture & Design

Understanding the idea behind modeling

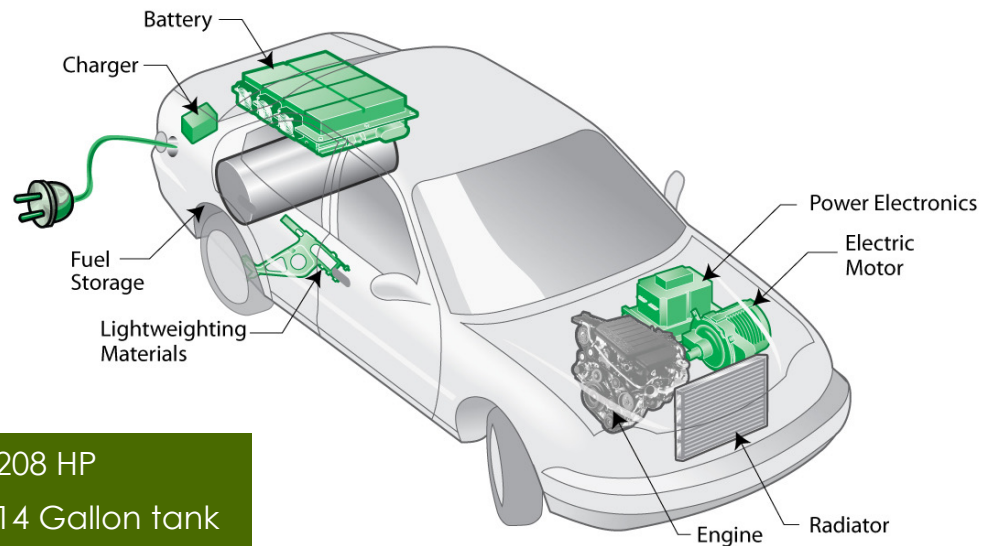


- 1 Conceptual Models
- 2 Architectural Models
- 3 Architecture Vs. Design

Conceptual Models



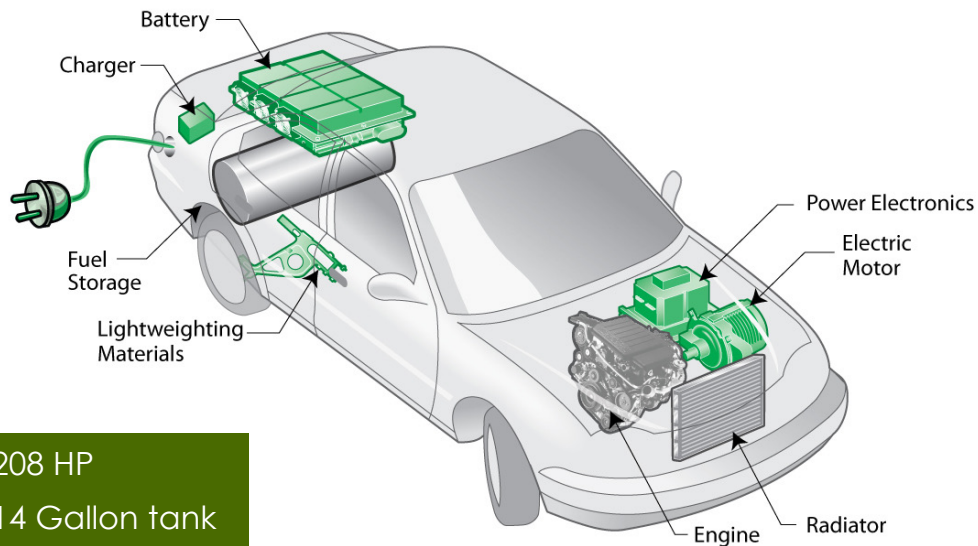
A representation of a system made from composition of concepts



208 HP
14 Gallon tank
70 kWh battery

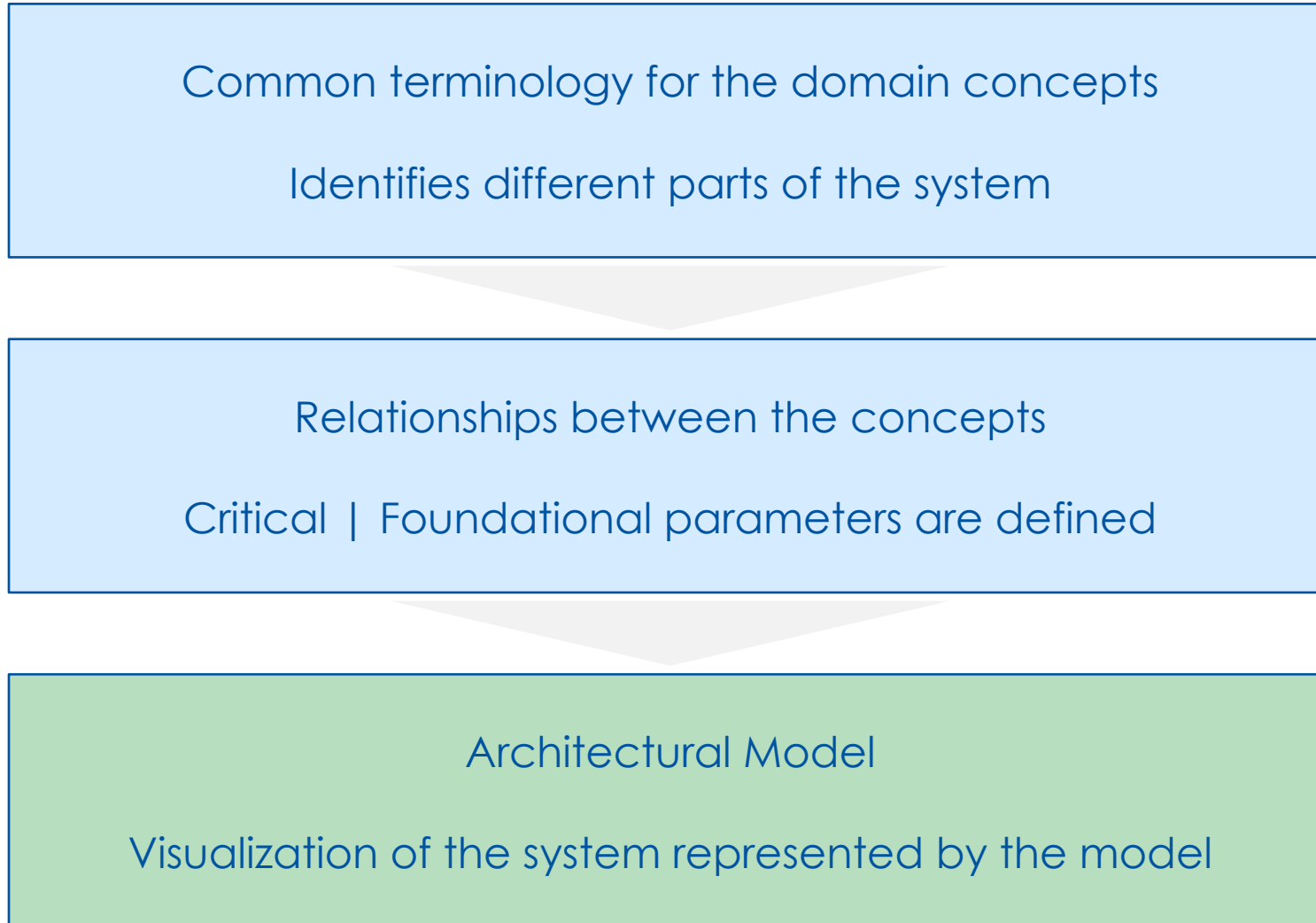
Objective of Conceptual Models

1. Enhance the understanding of the designers
2. Conveying the ideas to stakeholders
3. Provide a point of reference to create detailed specifications
4. Documentation for future reference



208 HP
14 Gallon tank
70 kWh battery

Software : Conceptual Models





Structured representation of a solution that meets the requirements in the problem space

- High level abstraction of parts of the end solution
- Presents a view of how the requirements will be met
- Assist in answering the questions posed by different stakeholders

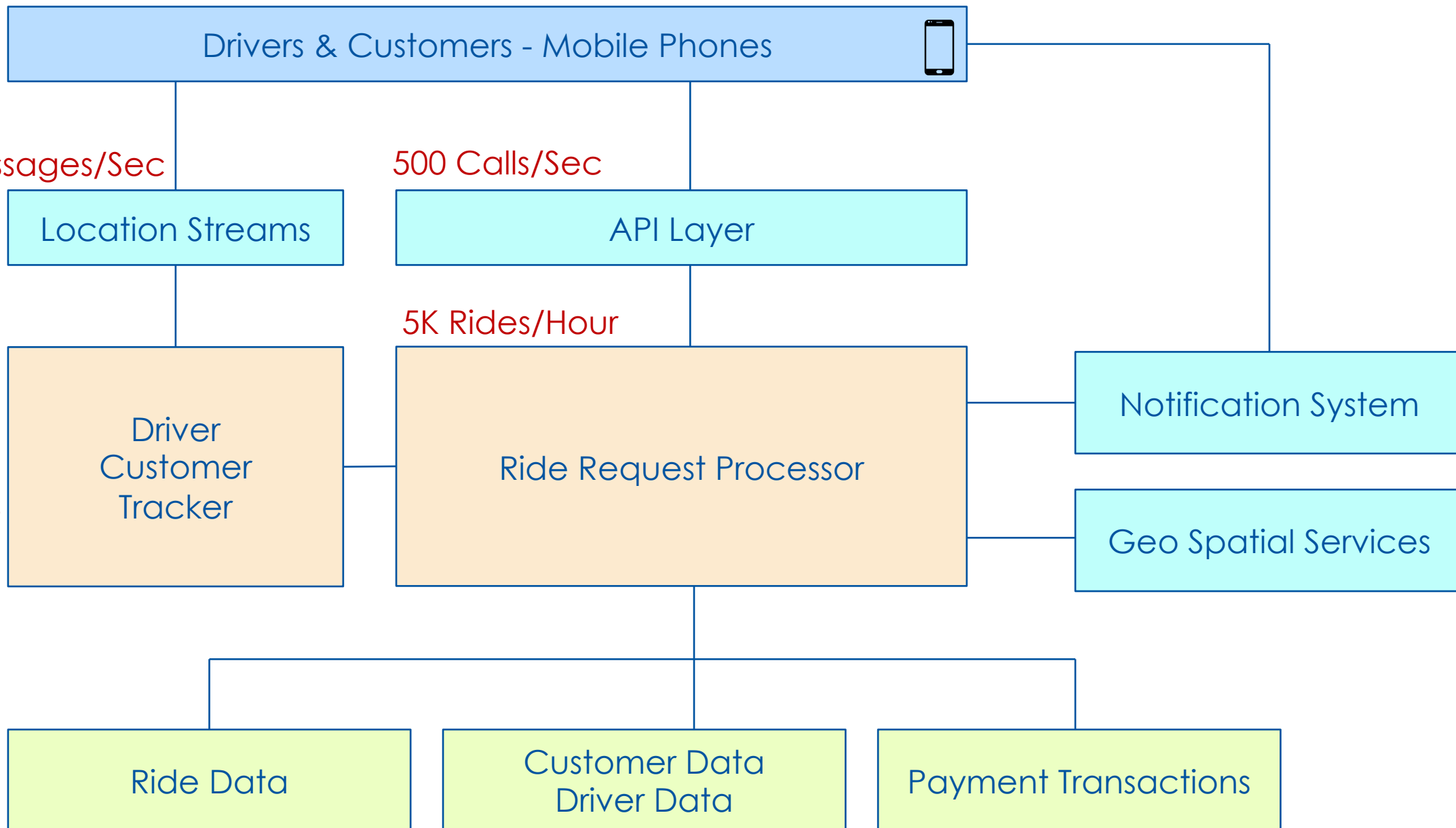
Architecture Vs. Design

Difference is in the level of details & focus

- **Architecture** = High Level | Skeleton | Long term focus
- **Design** = Relatively detailed | Focus on implementation

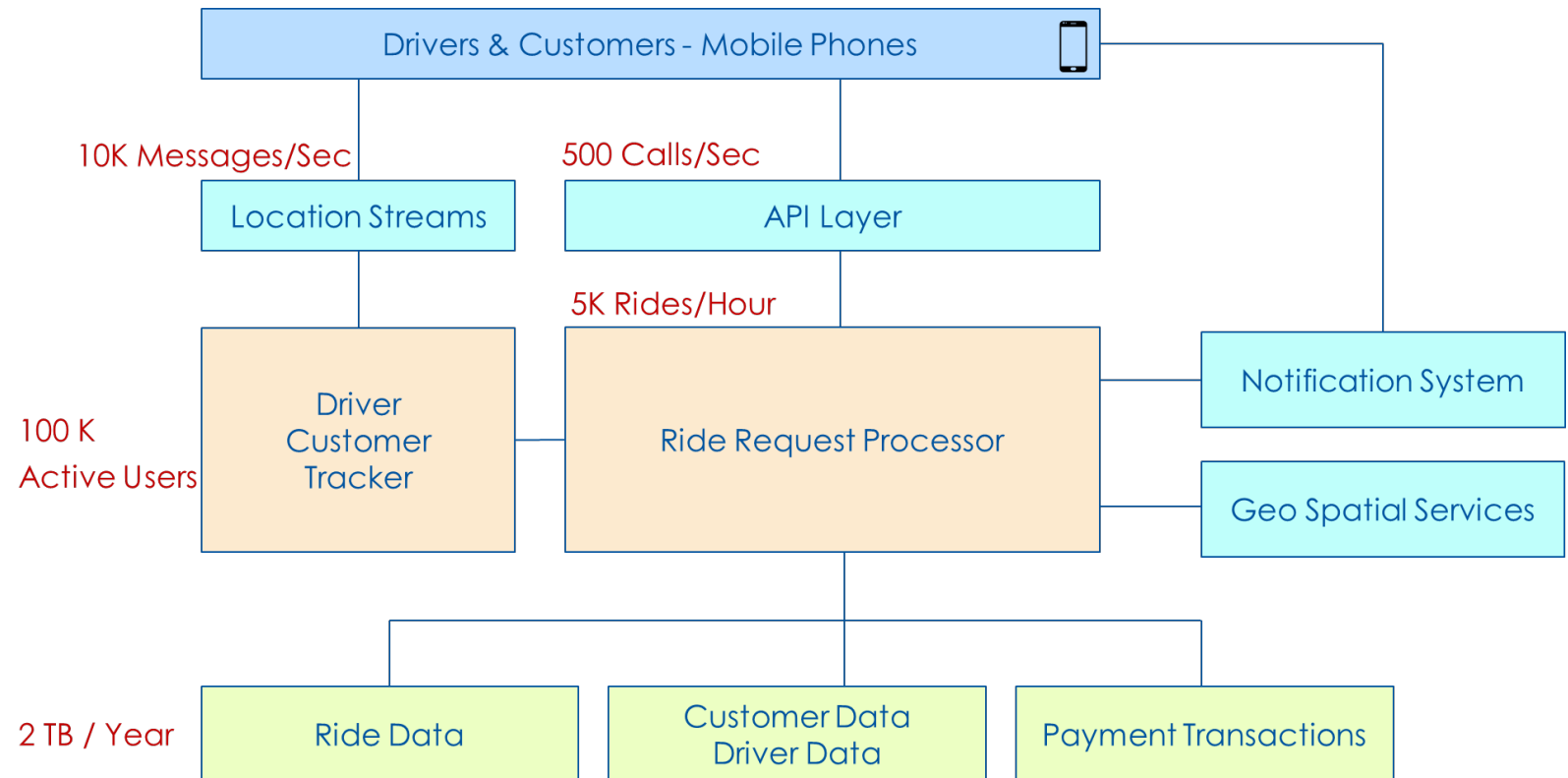
Architecture & Design for a Rideshare System





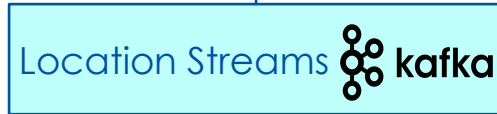
Quick Exercise

Think about the design

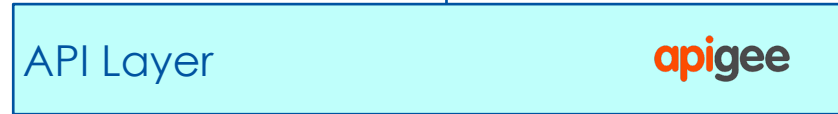




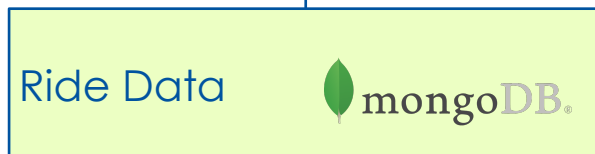
10K Messages/Sec



500 Calls/Sec



5K Rides/Hour



100 K
Active Users

2 TB / Year



Quick Review

Conceptual Model = A set of concepts & their relationships

Architectural Model = A structured representation of a solution

Design = A structured representation of a solution that is closer to the implementation

Modelling & Architecture style

Common Modeling Techniques and Architectural Styles



- 1 Modelling techniques
- 2 Architectural Styles
- 3 Intro to Domain Driven Design

Model Diagrams

Multiple ways of modelling

- Purpose
- Perspective | Viewpoint
- Level of Details

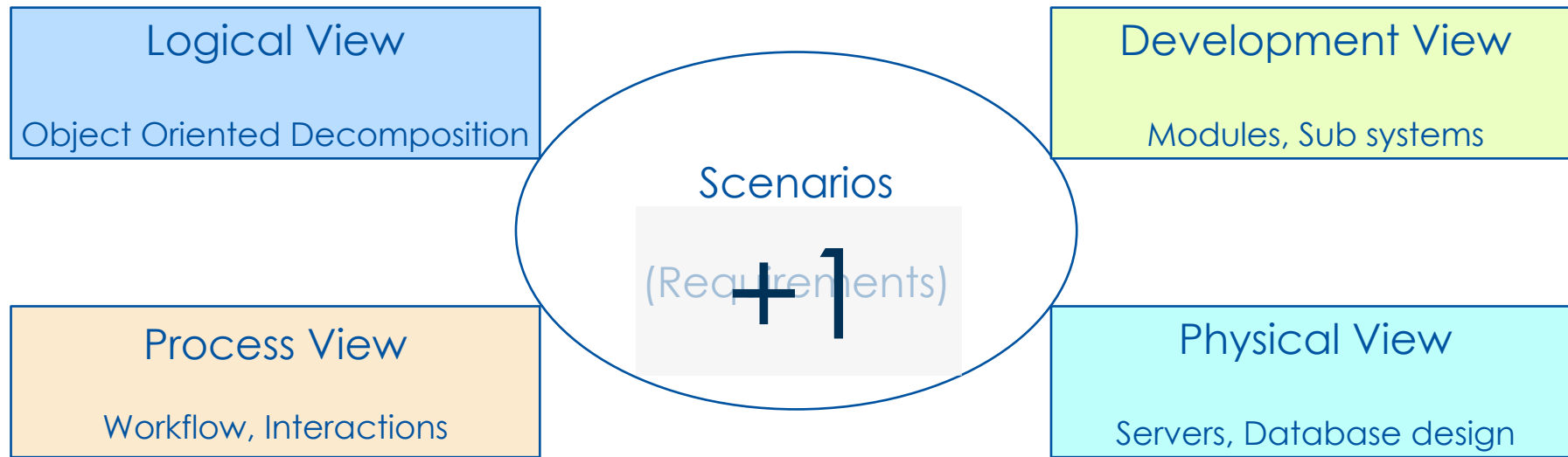
Models are NOT mutually exclusive i.e.; you may create multiple models for the same system

4+1 Architectural View Model

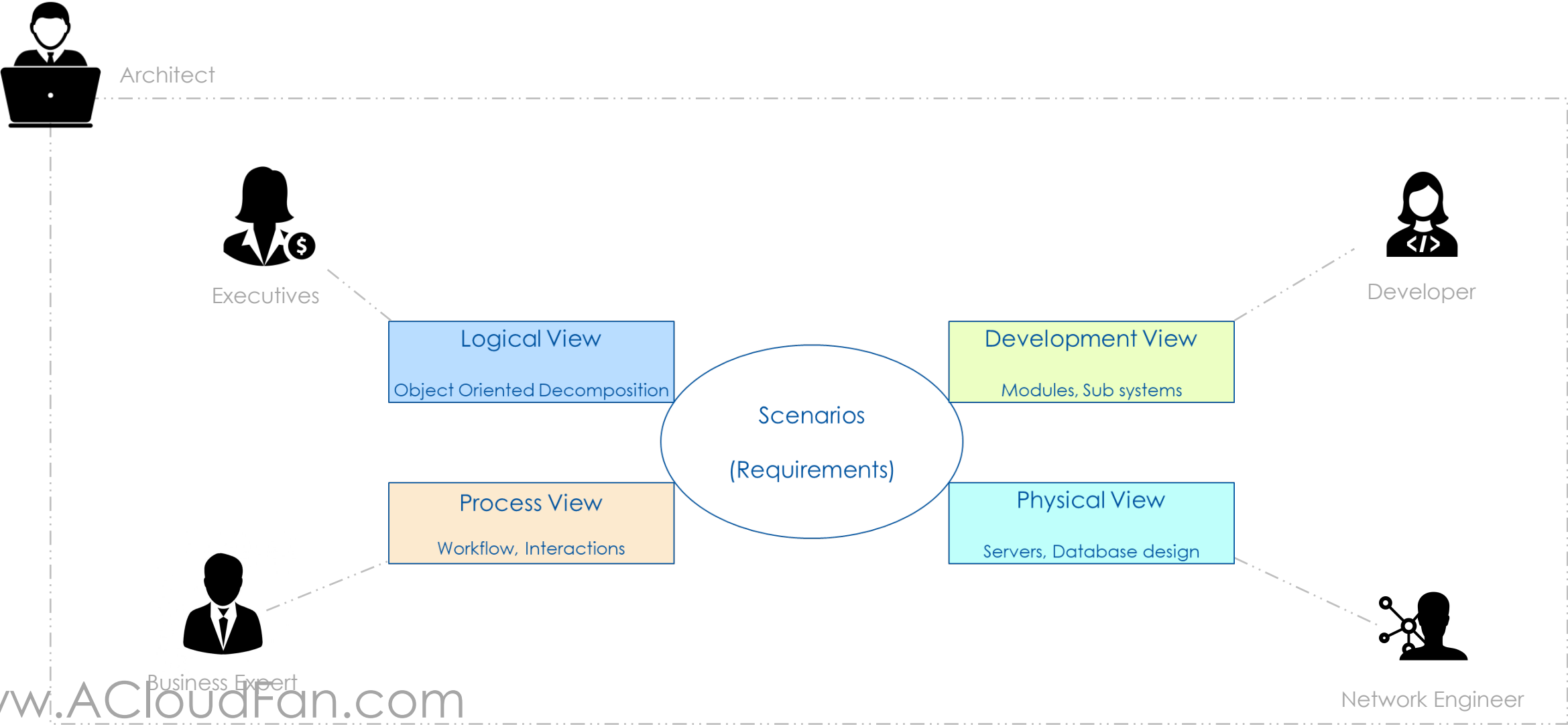


4 +1 Architectural View Model

Describe the architecture from viewpoints of multiple stakeholders

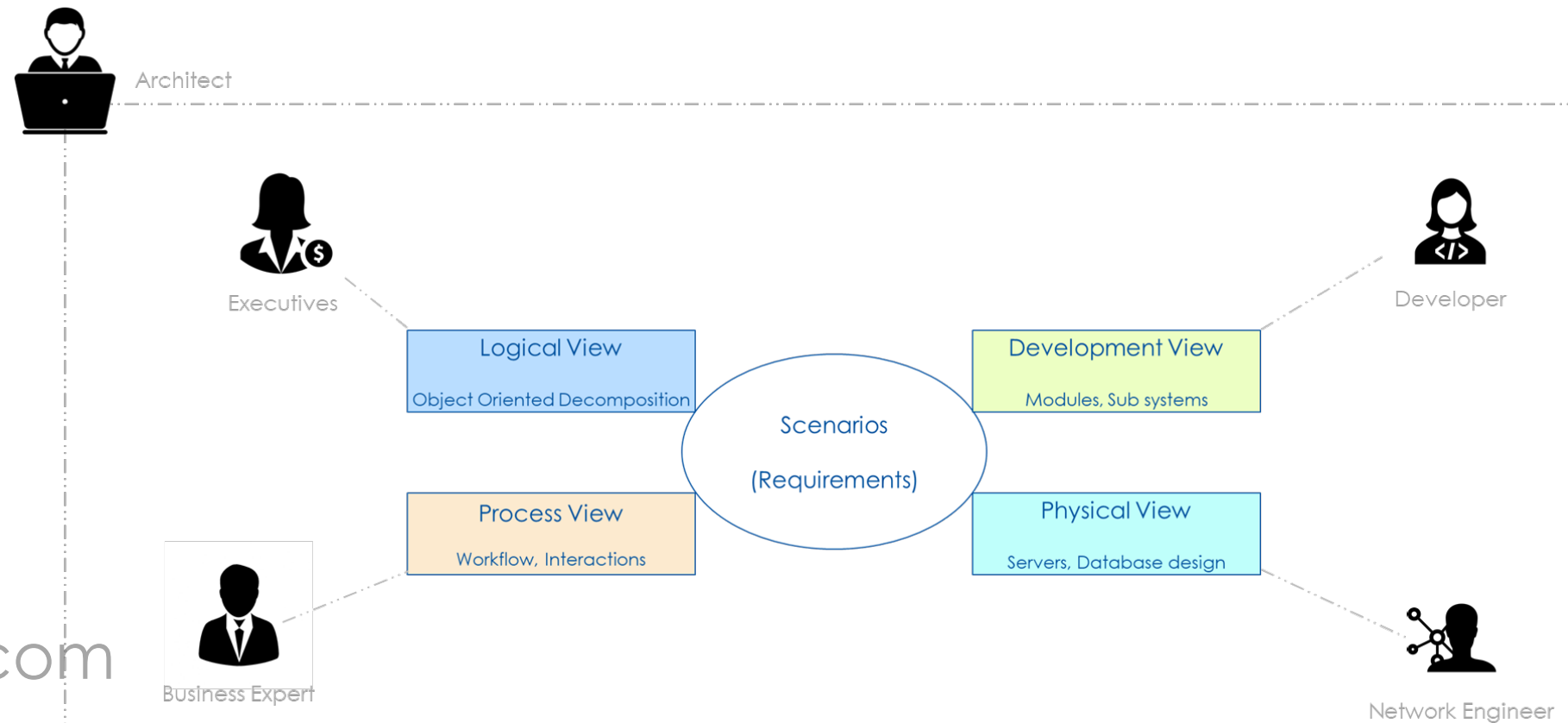


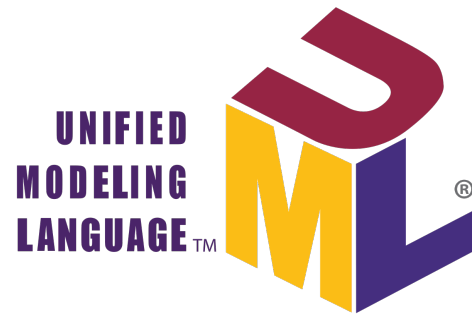
Stakeholder Interests | Concerns



Quick Exercise

List out the stakeholders in your organization





<http://www.omg.org/uml>

Please go through basic UML notation & diagrams on your own



<http://www.omg.org/uml>

A standard set of diagrams

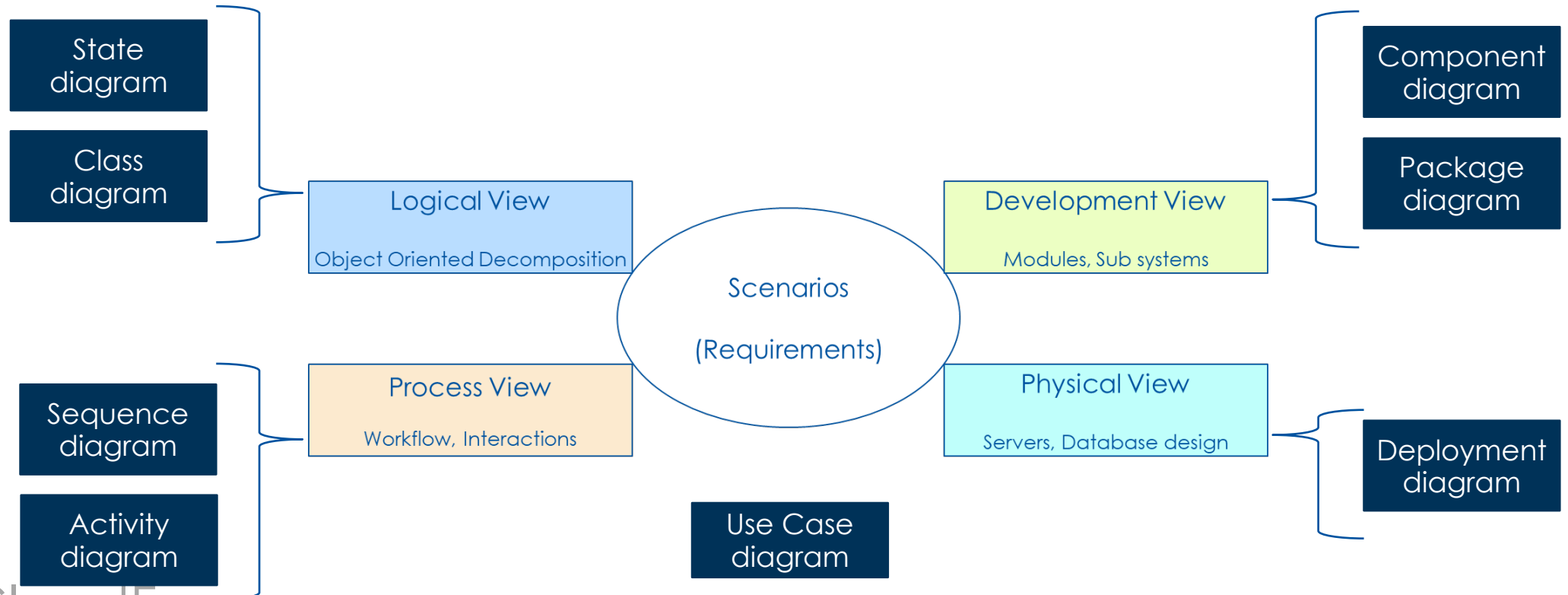
A standard set of notations

Please go through basic UML notation & diagrams on your own



<http://www.omg.org/uml>

A standard set of diagrams (14)



Software Architecture Styles



Reusable architectural pattern which may be used as a solution to a commonly occurring problem

Architectural Styles



Categorized based on the *Key Focus Area*

Communication

- Service Oriented Architecture (SOA)
- Message Bus Architecture

Structure

- Layered architectures
- Object Oriented Architecture & Design

Architectural Styles



Categorized based on the *Key Focus Area*

Deployment

- Client Servers
- 3 Tier Architecture

Data

- Database Centric Design
- Data Flow Diagrams

Architectural Styles



Categorized based on the *Key Focus Area*

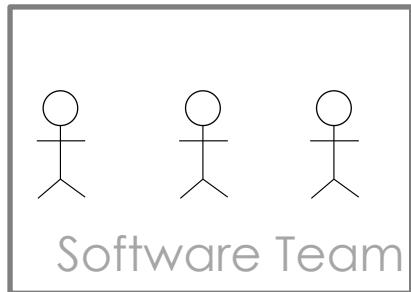
Domain

- Domain Driven Design

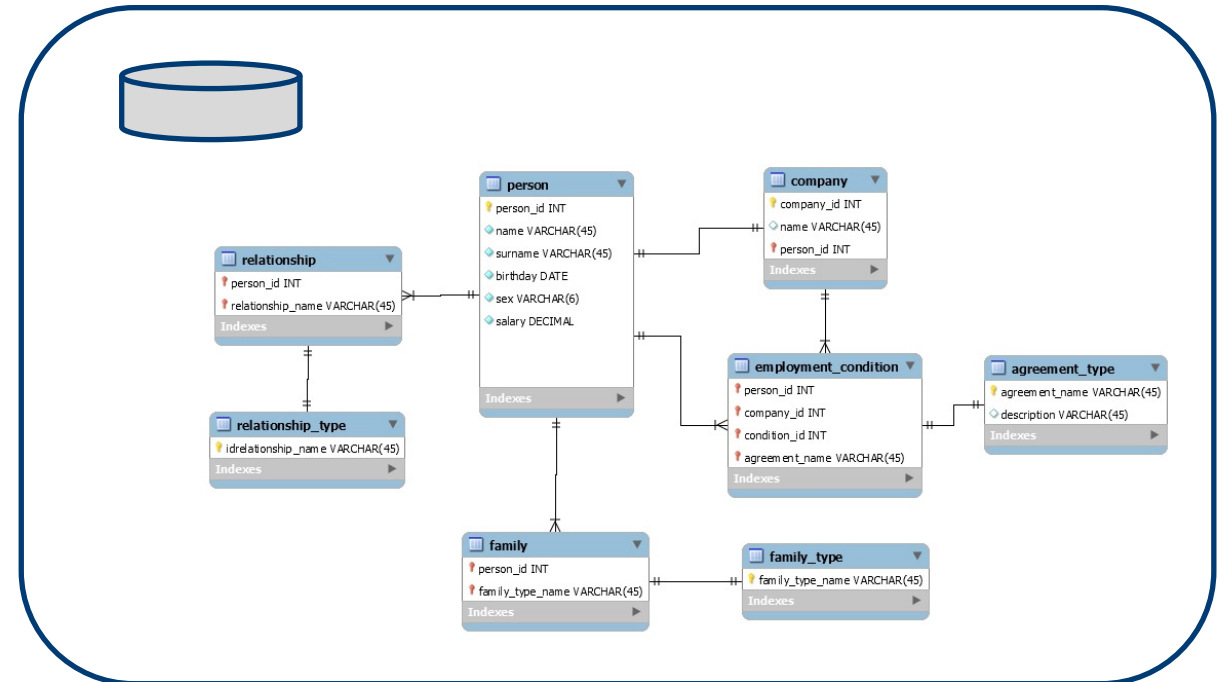
Focus is on Business Domain rather than technology

Data Centric Architecture

“ Focus is on core data in the domain



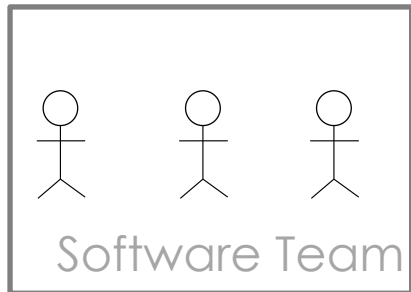
Develops



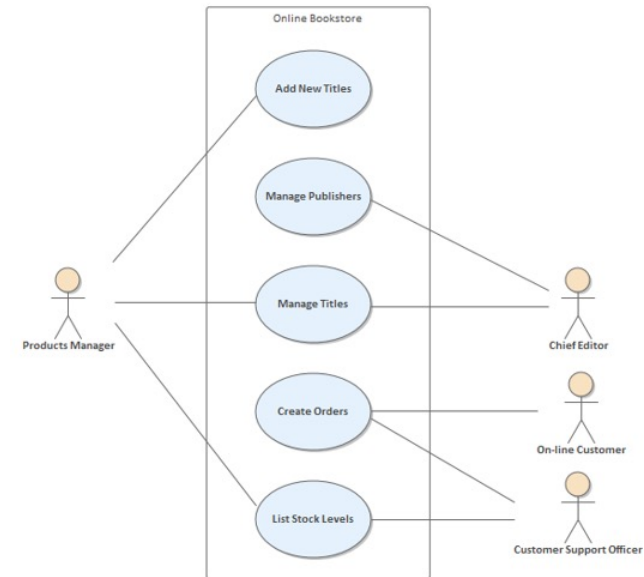
Process Centric Architecture



Focus is on core use cases in the business

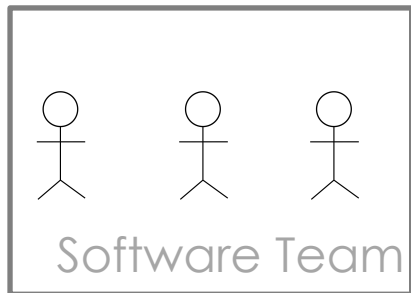


Develops

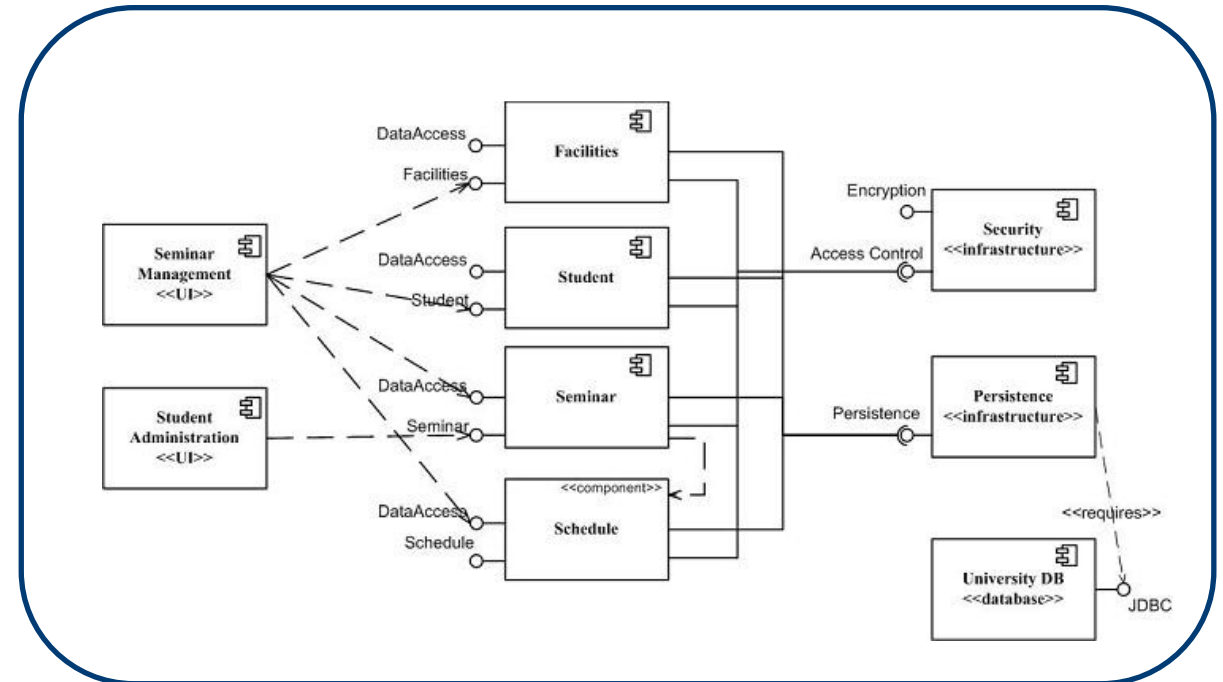


Services Centric Architecture (SOA | REST)

“ Focus is on core capabilities exposed as services

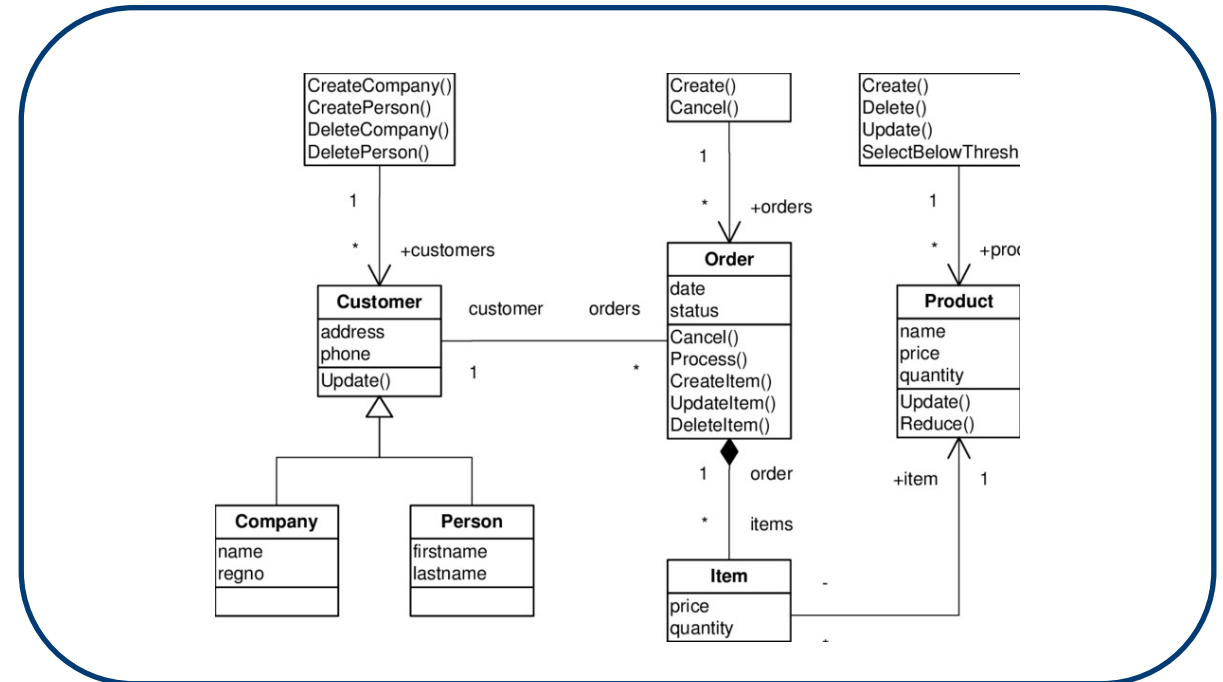
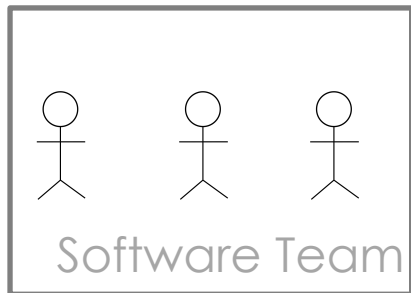


Develops



Object Oriented Architecture (OOA)

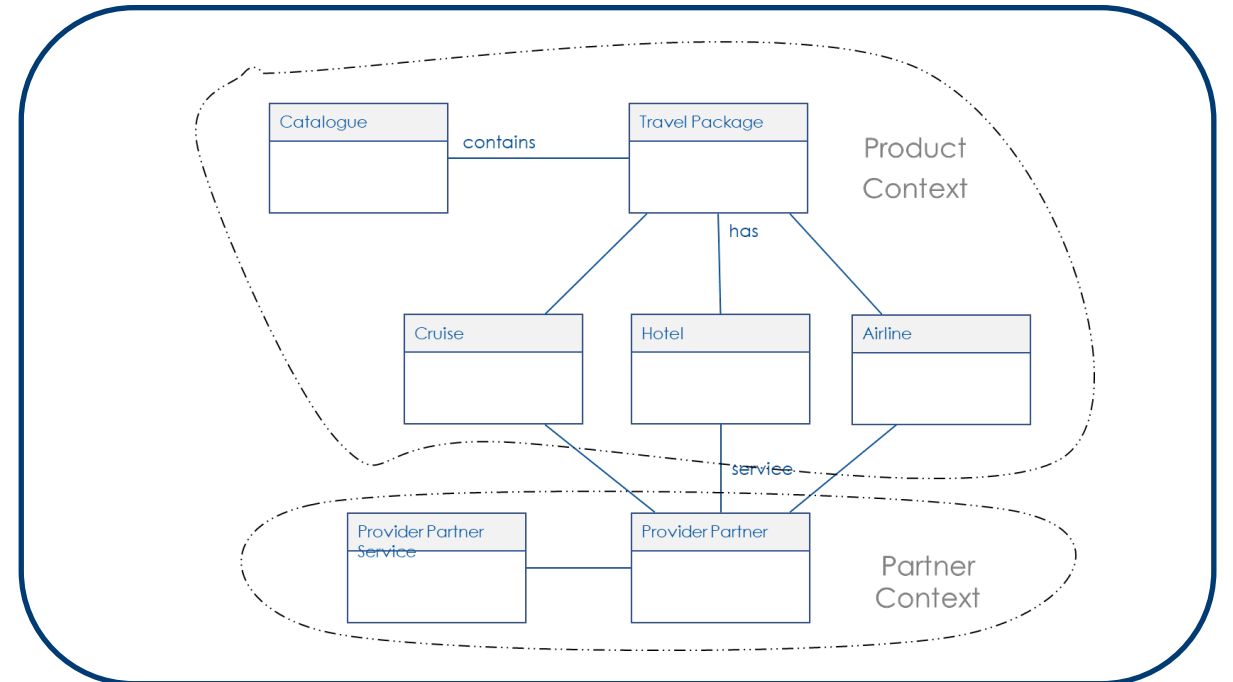
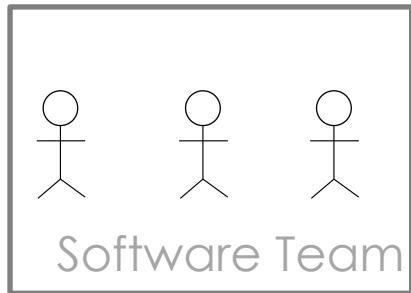
“ Focus is on identifying real work objects classes



Domain Driven Design



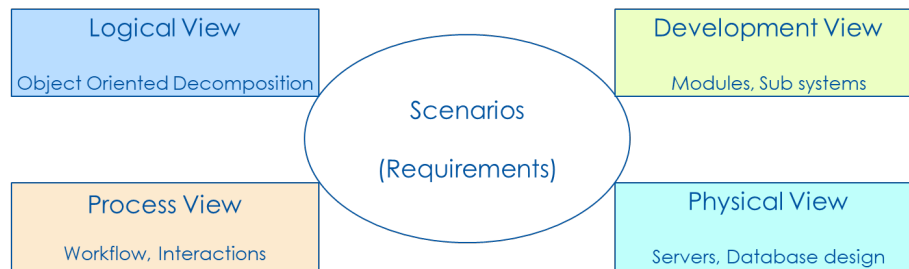
Focus is on the business domain





Review

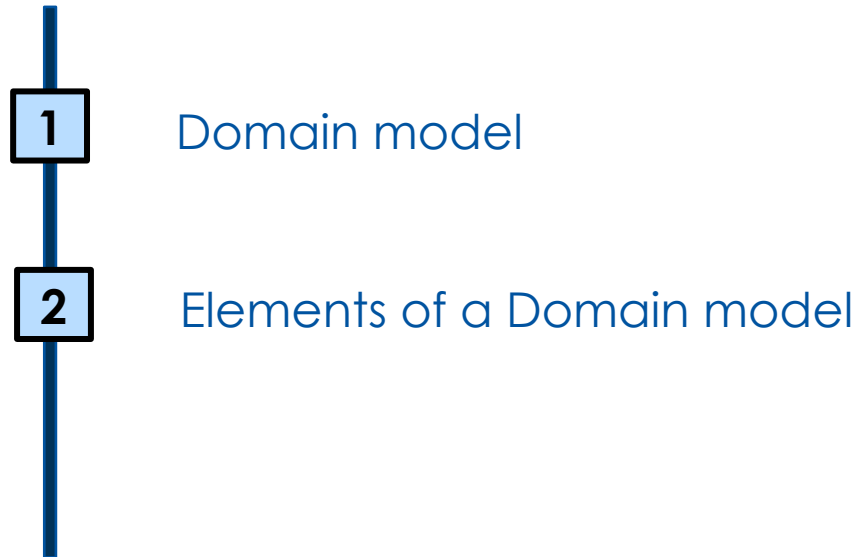
- Architects create models using different modeling techniques



- There are multiple architectural styles
 - Domain Driven Design = focusses on Business Domain

Domain Models

Understanding the business domain





WHY do businesses invest in software?

WHY do businesses invest in software?

- To fulfill some **need(s)** of the business | enterprise

Reduce manual labor

Increase Efficiency

Competitive Edge

... ..



WHY do businesses invest in software?

- To Solve some Business Problem(s)

Reduce manual labor

Reduce the size of the workforce to save \$\$\$!!

Increase Efficiency

Customer complaints about long response times !!

Competitive Edge

Competitors are pulling away customers !!

... ..

...

Business Problems



Current or long-term challenges & issues faced by the business that may prevent the business from achieving its goals.

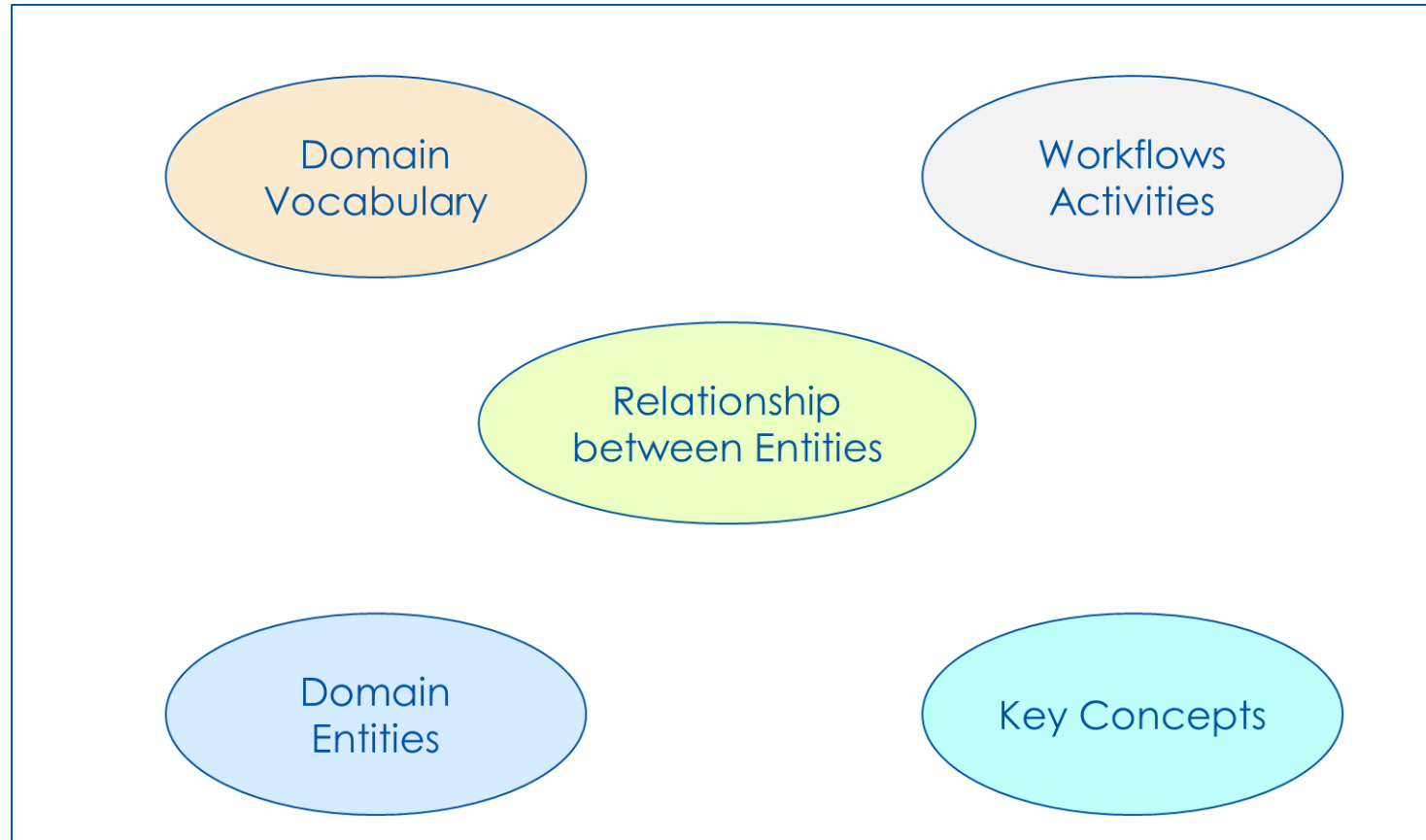
To understand the problem, architects **MUST** understand the domain first !!!

Domain Model



Organized and structured knowledge of the domain that is relevant for solving a business problem

Domain Model

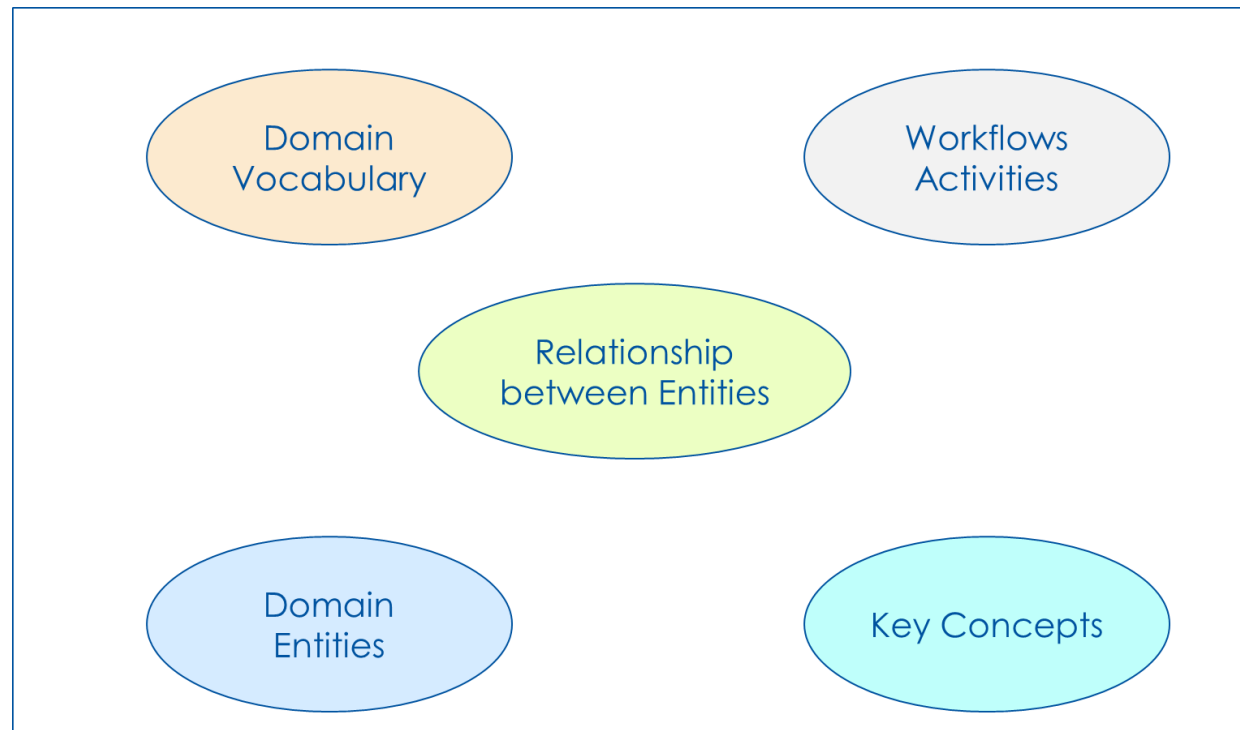


May contain additional knowledge



Review

- Domain Model = Structured Knowledge
 - Purpose = Solution for a Business Problem



Enterprise Domain Models

Also Known as "Aggregate" / "Unified" Domain Model



- 1 Knowledge crunching
- 2 Enterprise Domain Models
- 3 What challenges does DDD address?



**WHO in your Organization REALLY understands your
Org's Business Processes?**

Domain Experts

a.k.a. Subject Matter Experts or SME

Complex Domains - NO one expert knows everything about the domain !!!

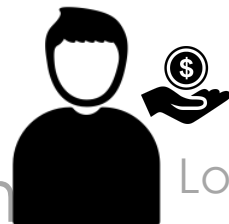
Banking



Retail Accounts Expert



Merchant Accounts Expert



Loans Accounts Expert



Compliance & Regulatory

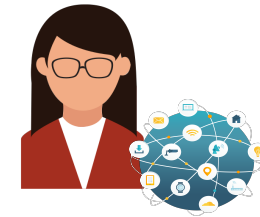
Domain Experts

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Travel Advisor



Partner Contracts



Accounts



Customer Support

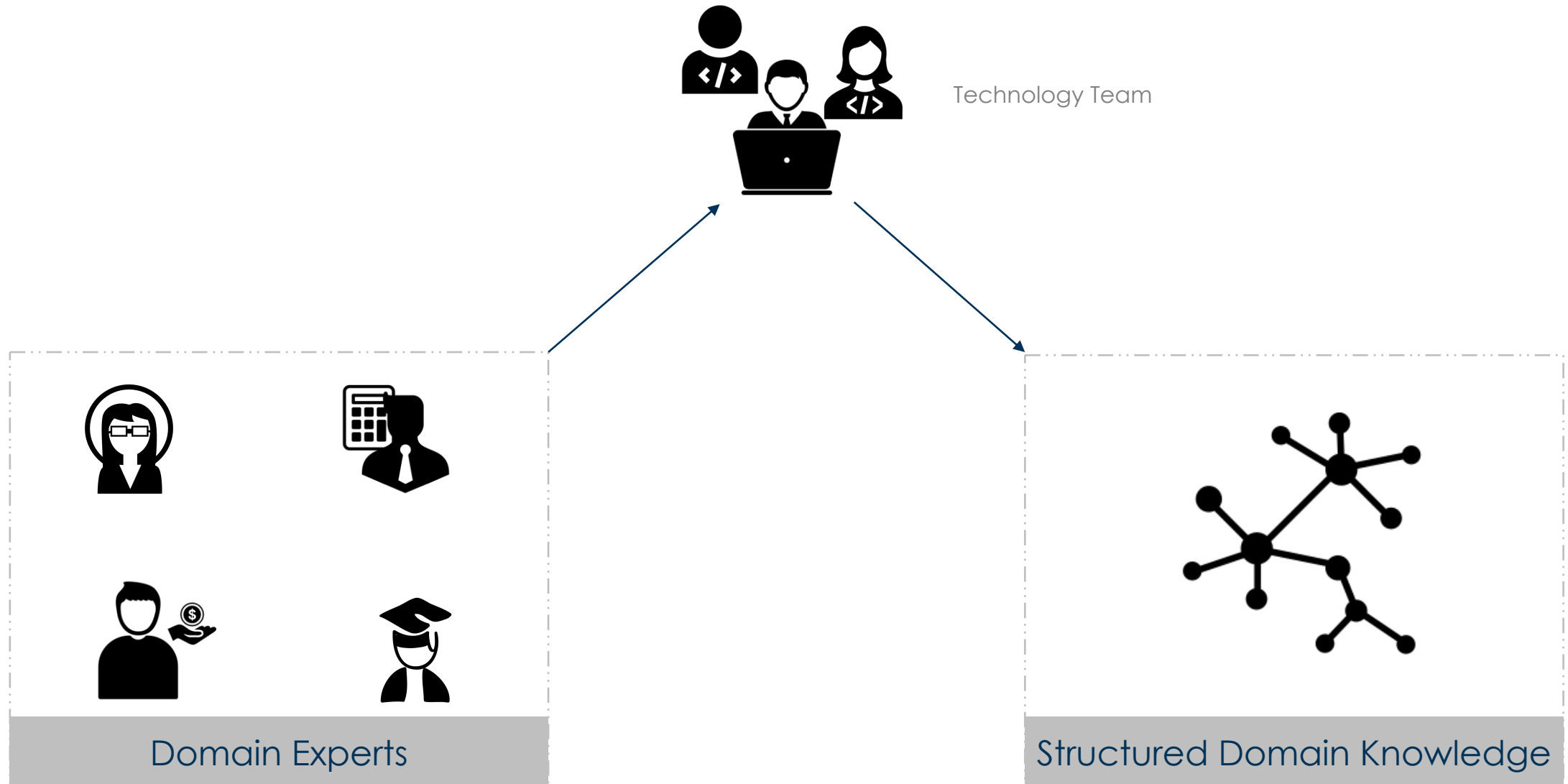
Knowledge Crunching



Teams process the knowledge received from the Domain Experts into domain models

This process is referred to as Knowledge Crunching

Knowledge Crunching

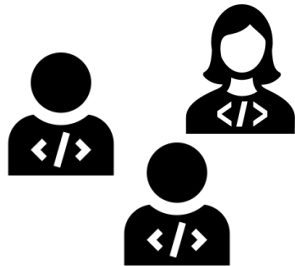


Technology Team

Led by an experienced technologist

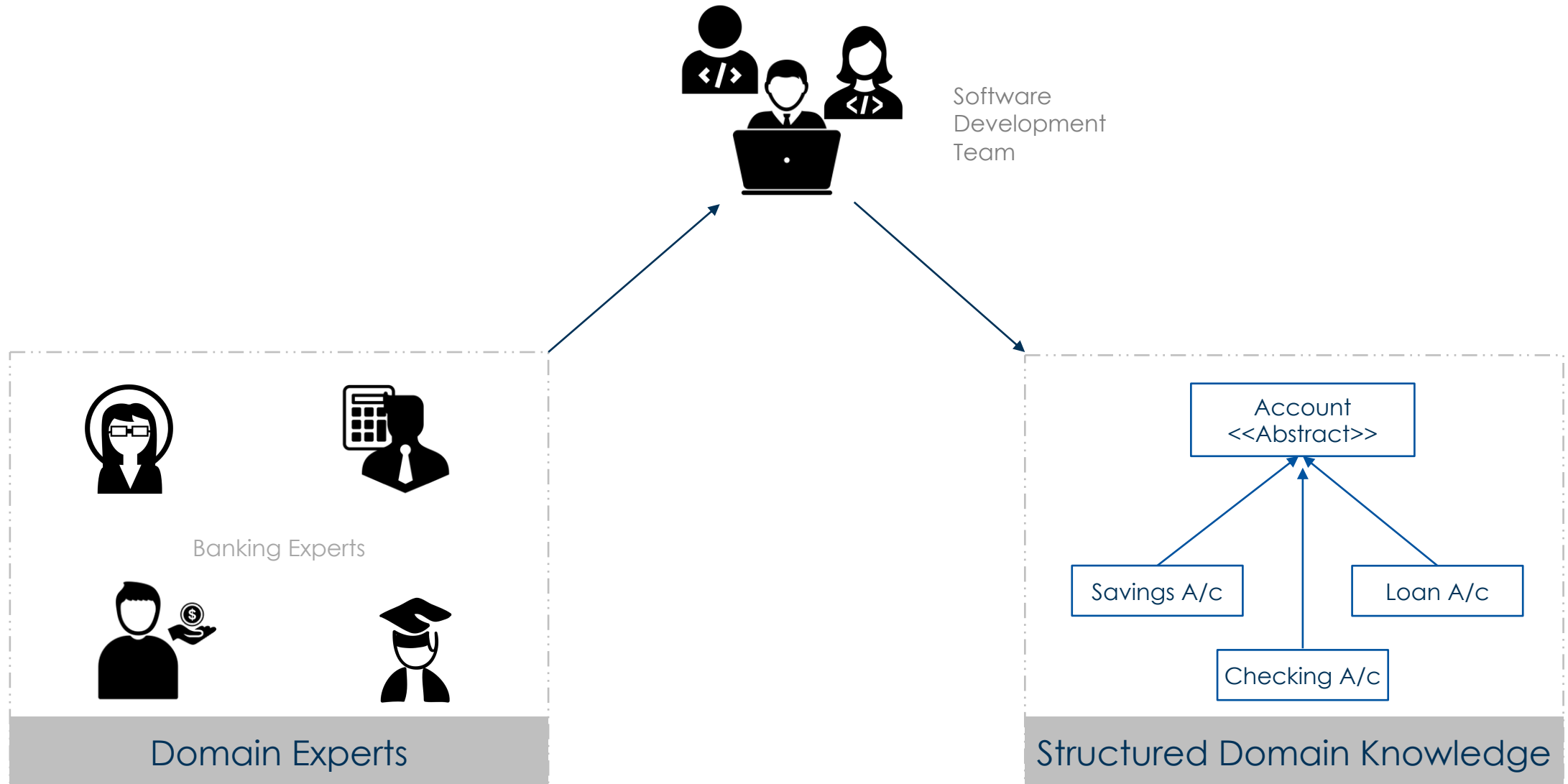


- IT Lead e.g., Architect, Lead Developer



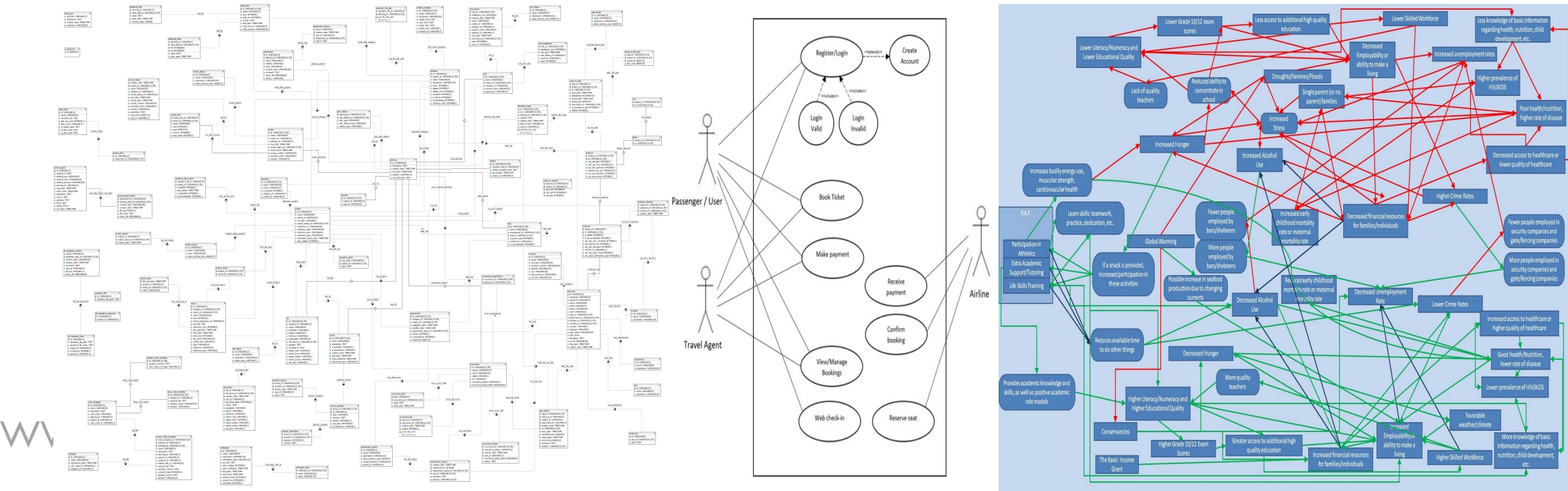
- Team members e.g., Developers, Analyst

Knowledge Crunching



Enterprise Models

Knowledge was gathered from multiple experts to create an enterprise domain model

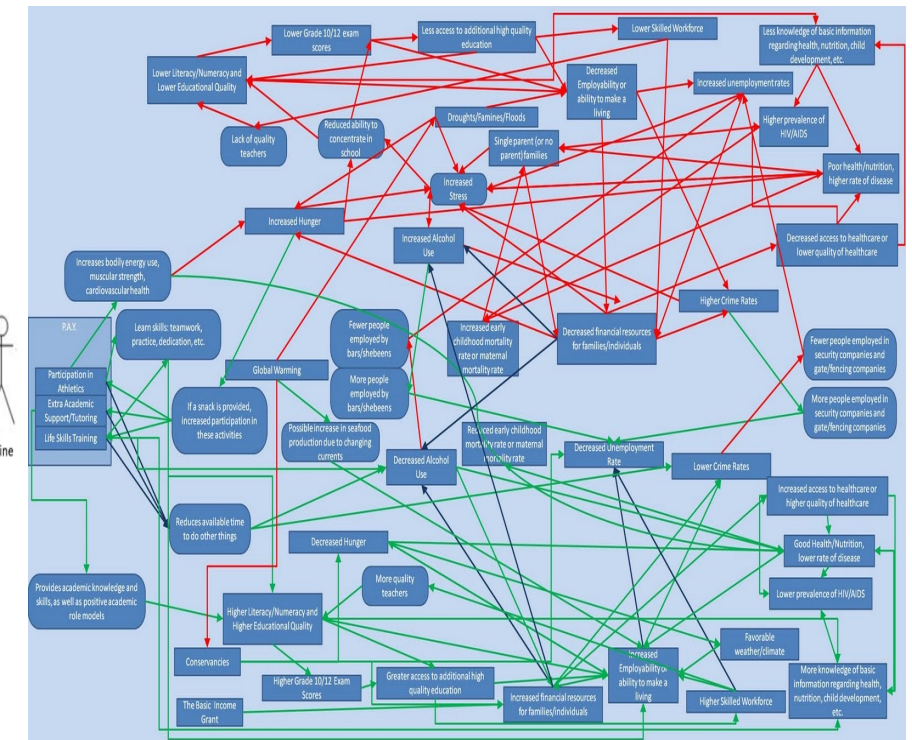
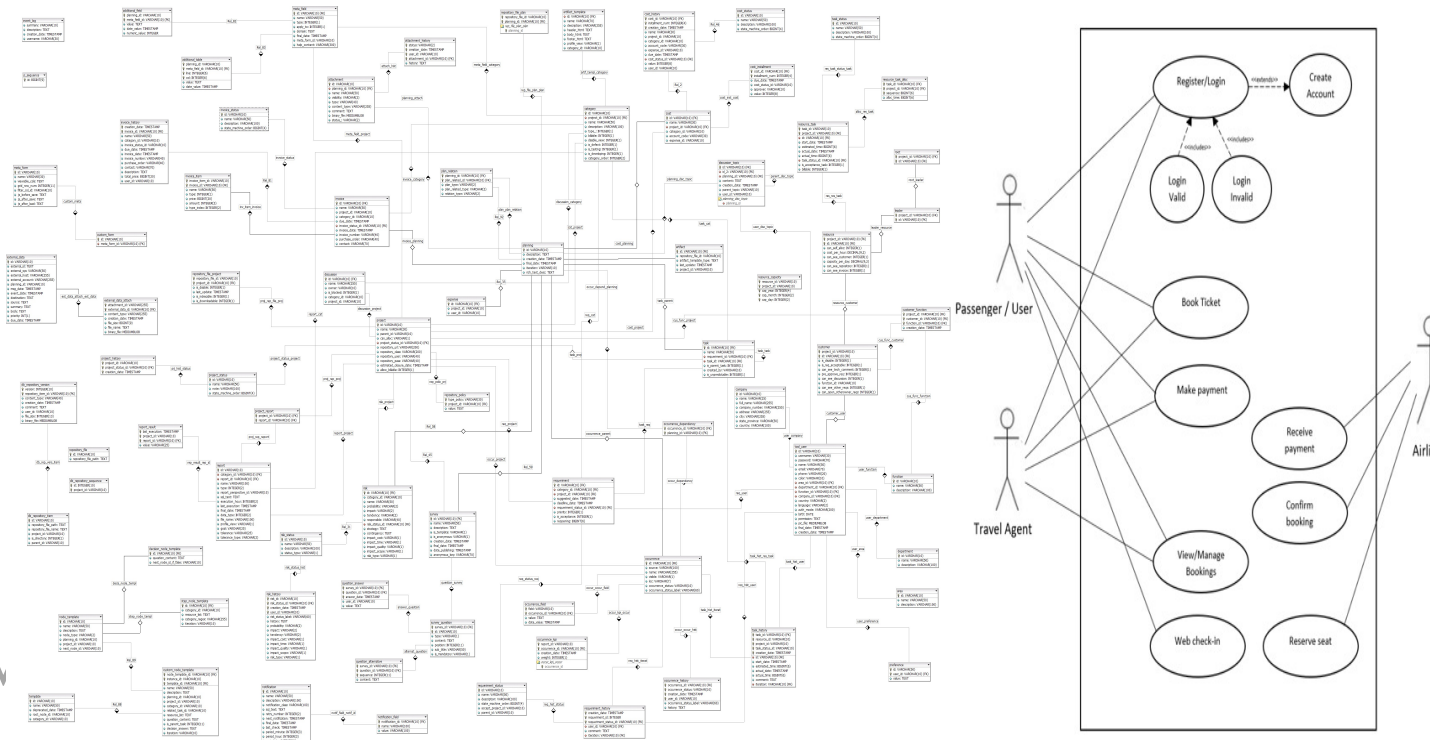


Enterprise Models

Also known as Unified Models

Also known as Canonical Models

Also known as Aggregate Models



Challenges with Unified models

1 Complexity

2 Ownership

3 Linguistic



Complexity

Inherent complexity due to scope and size

Overlapping & Redundant capabilities



Ownership

No single expert who own's the entire model

Model falls behind the reality and over time loses its value

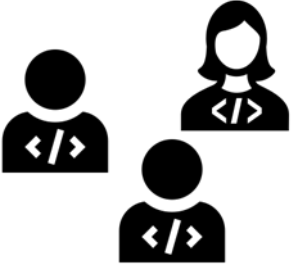
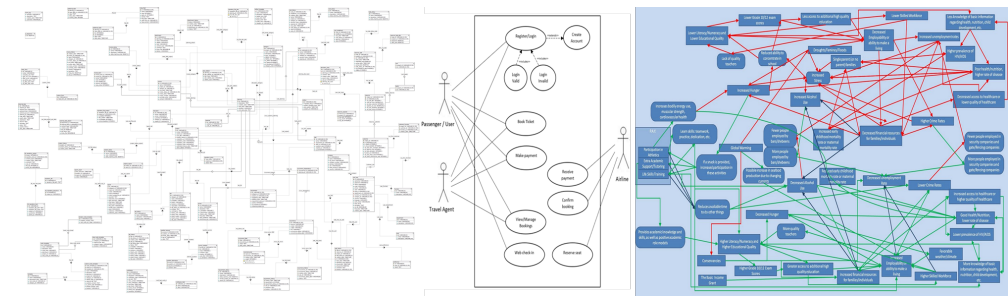


Linguistic

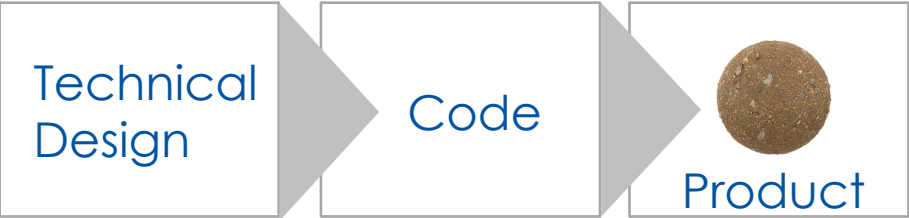
Same business term has different meanings in different domains

Technology teams | Business teams speak different languages

Software Development



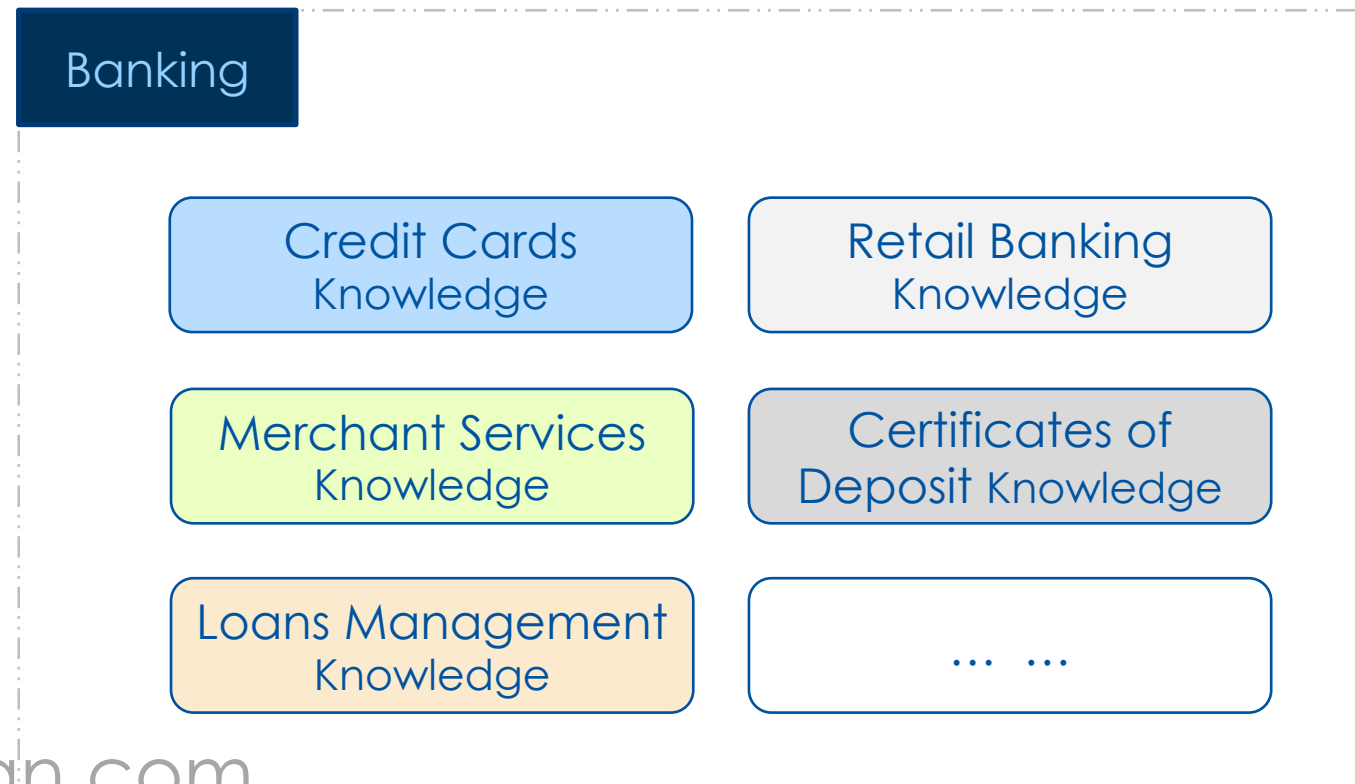
Software
Development
Team



Aggregate Domain Models

a.k.a. Holistic/Enterprise Domain Model

A domain model that covers all the facets of the domain



Addressing the Challenges



Domain Driven Design approach provides principles and patterns to address the challenges faced with developing complex domain models



Review

- Knowledge Crunching = Creating the Domain Model
- Multiple challenges with creating models for complex domains
- DDD addresses these challenges